

Link:

<https://drive.google.com/file/d/1xC DuaWq21GQRXkobXkan6PW9oJyiIMuO/view?usp=sharing>

Código:

```
1  #include <WiFi.h>
2  #include <WebServer.h>
3  #include <LiquidCrystal.h>
4
5  // ----- CONFIG WiFi -----
6  const char* WIFI_SSID = "UPCH_CENTRAL"; // Nombre exacto de la red Wi-Fi
7  const char* WIFI_PASS = "CAYETANO2022"; // Contraseña
8
9
10 // ----- LCD paralelo 4 bits -----
11 LiquidCrystal lcd(21, 14, 16, 17, 18, 19); // RS, EN, D4, D5, D6, D7
12
13 // ----- HTTP -----
14 WebServer server(80);
15
16 // Decodifica %xx y convierte '+' en espacio
17 String urlDecode(const String& s) {
18     String out;
19     out.reserve(s.length());
20     for (size_t i = 0; i < s.length(); i++) {
21         char c = s[i];
22         if (c == '+') { out += ' '; }
23         else if (c == '%' && i + 2 < s.length()) {
24             char h1 = s[i+1], h2 = s[i+2];
25             auto hex2int = [](char h)->int {
26                 if (h >= '0' && h <= '9') return h - '0';
27                 if (h >= 'a' && h <= 'f') return 10 + (h - 'a');
28                 if (h >= 'A' && h <= 'F') return 10 + (h - 'A');
29                 return 0;
30             };
31             int v = (hex2int(h1) << 4) | hex2int(h2);
32             out += char(v);
33             i += 2;
34         } else {
35             out += c;
```

```
36     }
37 }
38 return out;
39 }
40
41 void handleRoot() {
42     server.send(200, "text/plain",
43         "ESP32 LCD OK. Use /lcd?msg=TU_TEXTO");
44 }
45
46 void handleLCD() {
47     if (!server.hasArg("msg")) {
48         server.send(400, "text/plain", "Falta parametro msg");
49         return;
50     }
51
52     String msg = urlDecode(server.arg("msg"));
53
54     // Limita a 16 chars para la 2da linea (evita desbordes)
55     if (msg.length() > 16) msg = msg.substring(0, 16);
56
57     // Actualiza LCD sin parpadeo
58     lcd.setCursor(0, 0);
59     lcd.print("Mensaje recibido ");
60     // limpia la linea 2
61     lcd.setCursor(0, 1);
62     lcd.print("          ");
63     lcd.setCursor(0, 1);
64     lcd.print(msg);
65
66     // Respuesta a la app
67     server.send(200, "text/plain", "OK: " + msg);
68 }
69
```

```
70 void setup() {
71     Serial.begin(115200);
72
73     // LCD
74     lcd.begin(16, 2);
75     lcd.print("Iniciando WiFi...");
76
77     // WiFi
78     Wifi.mode(WIFI_STA);
79     Wifi.begin(WIFI_SSID, WIFI_PASS);
80
81     Serial.print("Conectando a WiFi");
82     while (Wifi.status() != WL_CONNECTED) {
83         delay(500);
84         Serial.print(".");
85     }
86     Serial.println();
87     Serial.print("Conectado. IP: ");
88     Serial.println(Wifi.localIP());
89
90     lcd.clear();
91     lcd.setCursor(0, 0);
92     lcd.print("WiFi conectado");
93     lcd.setCursor(0, 1);
94     lcd.print(Wifi.localIP().toString());
95
96     // Rutas HTTP
97     server.on("/", handleRoot);
98     server.on("/lcd", handleLCD);
99     server.begin();
100     Serial.println("Servidor HTTP listo");
101 }
```

```

77 // WiFi
78 WiFi.mode(WIFI_STA);
79 WiFi.begin(WIFI_SSID, WIFI_PASS);
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96 // Rutas HTTP
97 server.on("/", handleRoot);
98 server.on("/lcd", handleLCD);
99 server.begin();
100 Serial.println("Servidor HTTP listo");
101 }
102
103 void loop() {
104     server.handleClient();
105 }

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#include <WebServer.h>
#include <LiquidCrystal.h>

```

```

// ----- CONFIG WiFi -----

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// ----- LCD paralelo 4 bits -----  
LiquidCrystal lcd(21, 14, 16, 17, 18, 19); // RS, EN, D4, D5, D6, D7
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// ----- HTTP -----  
WebServer server(80);
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```
// Decodifica %xx y convierte '+' en espacio  
String urlDecode(const String& s) {  
    String out;  
    out.reserve(s.length());  
    for (size_t i = 0; i < s.length(); i++) {  
        char c = s[i];  
        if (c == '+') { out += ' '; }  
        else if (c == '%' && i + 2 < s.length()) {  
            char h1 = s[i+1], h2 = s[i+2];  
            auto hex2int = [](char h)->int {  
                if (h >= '0' && h <= '9') return h - '0';  
                if (h >= 'a' && h <= 'f') return 10 + (h - 'a');  
                if (h >= 'A' && h <= 'F') return 10 + (h - 'A');  
                return 0;  
            };  
            int v = (hex2int(h1) << 4) | hex2int(h2);  
            out += char(v);  
            i += 2;  
        } else {  
            out += c;  
        }  
    }  
    return out;  
}
```

```
void handleRoot() {  
    server.send(200, "text/plain",  
        "ESP32 LCD OK. Use /lcd?msg=TU_TEXTO");  
}
```

```
void handleLCD() {  
    if (!server.hasArg("msg")) {  
        server.send(400, "text/plain", "Falta parametro msg");  
        return;  
    }  
}
```

```
String msg = urlDecode(server.arg("msg"));
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```
// Limita a 16 chars para la 2da linea (evita desbordes)  
if (msg.length() > 16) msg = msg.substring(0, 16);
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// Actualiza LCD sin parpadeo
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```

    lcd.setCursor(0, 0);
    lcd.print("Mensaje recibido ");
    // limpia la linea 2
    lcd.setCursor(0, 1);
    lcd.print("          ");
    lcd.setCursor(0, 1);
    lcd.print(msg);

    // Respuesta a la app
    server.send(200, "text/plain", "OK: " + msg);
}

void setup() {
    Serial.begin(115200);

    // LCD
    lcd.begin(16, 2);
    lcd.print("Iniciando WiFi...");

    // WiFi
    WiFi.mode(WIFI_STA);
    WiFi.begin(WIFI_SSID, WIFI_PASS);

    Serial.print("Conectando a WiFi");
    while (WiFi.status() != WL_CONNECTED) {
        delay(500);
        Serial.print(".");
    }
    Serial.println();
    Serial.print("Conectado. IP: ");
    Serial.println(WiFi.localIP());

    lcd.clear();
    lcd.setCursor(0, 0);
    lcd.print("WiFi conectado");
    lcd.setCursor(0, 1);
    lcd.print(WiFi.localIP().toString());

    // Rutas HTTP
    server.on("/", handleRoot);
    server.on("/lcd", handleLCD);
    server.begin();
    Serial.println("Servidor HTTP listo");
}

void loop() {
    server.handleClient();
}

```